

Research Gap Identification — Literature Review Dossier

Evidence-grounded reduction of hallucination in LLM verification systems

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1. Final Research Topic

Evidence-grounded reduction of hallucination in large-language-model verification systems: from soft factuality checks to a structurally-verified, multi-agent claim-checking gate.

Focused on: (a) *problem* — hallucination / unsupported assertions in LLM output; (b) *mechanism* — evidence grounding, from soft prompt-based grounding to a hard, mechanically-verified evidence gate; (c) *setting* — automated claim/fact verification. This is the literature space of the Aletheia system.

2. Keywords

large language models; hallucination detection; hallucination mitigation; retrieval-augmented generation; evidence grounding; factual consistency; automated claim / fact verification; multi-agent LLM systems; faithfulness; misinformation detection

3. Literature Review Table (35 original empirical journal articles)

Reviews, systematic reviews, meta-analyses, and conference/proceedings papers deliberately excluded. Rows marked (R) were read from source; rows marked (C) are real journal articles whose author/details you must confirm from the DOI.

#	Author & Year	Paper / Journal	Objective	Sample / Area	Method	Key Findings	Limitations
1	Massenon 2025 (R)	LLM hallucinations in app reviews / Sci. Reports	Taxonomy of real LLM hallucinations	3M reviews, 90 apps	NMF + VADER + manual annot.	Factual errors 38% of reports	English-only reviews
2	Chelli 2024 (C)	Hallucination rates of ChatGPT/Bard / JMIR	Measure hallucinated references	Rotator-cuff review refs	Comparative ref. analysis	Many hallucinated references	Single clinical topic
3	npj Dig. Med. 2025 (C)	Clinical-safety hallucination framework / npj Dig. Med.	Quantify safety risk of LLM summaries	12,999 clinician sentences	Clinician-annotation framework	1.47% hallucination, 3.45% omission	Summarisation only

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4	NEJM AI 2025 (C)	Verify facts in LLM clinical notes via EHR / NEJM AI	Verify LLM text against records	Care docs + EHR	Fact-check vs EHR	Detectable factual errors	Single-site EHR
5	Yanagita 2023 (R)	ChatGPT on Japanese medical exam / JMIR Form. Res.	LLM factual accuracy, non-English	292 NMLE questions	GPT-3.5 vs GPT-4 + physicians	GPT-4 81.5% vs 42.8%	Excluded image questions
6	Comput. Biol. Med. 2024 (C)	GPT-4 vs Gemini references / Comput. Biol. Med.	Compare citation credibility	Medical research refs	Paired t-tests	Differences in ref. reliability	Citations only
7	Khatrri 2025 (C)	ChatGPT drug-info accuracy / JACCP	Accuracy + reproducibility	Drug-info questions	Expert eval of repeats	Variable accuracy/reproducibility	Single domain
8	Krishnamurthy 2024 (C)	Credibility framework for fact-checking / IEEE Access	Add credibility layer to LLM checking	Claim sets	LLM + credibility scoring	Credibility signals help	Framework; limited eval
9	npj Dig. Med. 2025 (C)	RAG for 10 LLMs, medical fitness / npj Dig. Med.	Does RAG generalize across LLMs	10 LLMs, fitness assess.	RAG over guidelines	GPT-4+RAG 96.4%, no hallucination	Single domain
10	npj Dig. Med. 2025 (C)	Long-context RAG for medical QA / npj Dig. Med.	Long-context grounded QA	Medical QA	Long-context RAG	Longer context improves faithfulness	Medical only
11	npj Dig. Med. 2025 (C)	RAG for local LLM in radiology / npj Dig. Med.	Improve small-LLM factuality	Radiology queries	RAG over guidelines	RAG lifts local-LLM accuracy	Single specialty
12	JMIR 2025 (C)	Two-layer RAG, low-resource medical QA / JMIR	Grounded QA with scarce corpora	Reddit medical Q&A	Two-layer RAG	Improves low-resource grounding	Proof-of-concept
13	J. Intell. Inf. Syst. 2026 (C)	Multimodal misinfo via RAG+LLM / JIIS	Cross-language RAG detection	Multilingual multimodal	RAG + LLM verifier	RAG improves cross-lingual detection	Evidence quality varies
14	ESWA 2024 (C)	Factual-consistency eval in LLM era / ESWA	Better factual-consistency metric	Summ. factuality sets	LLM-based metric	Better human correlation	Summarisation only
15	ESWA 2025 (C)	Multi-agent collaborative filtering / ESWA	Reduce reasoning hallucination	Reasoning/QA sets	Multi-agent + filtering	Cross-checking lowers hallucination	Compute-heavy; not gated

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16	Martin 2022 (C)	FacTeR-Check (similarity + NLI) / Knowl.-Based Syst.	Evidence retrieval + verification	COVID Spanish claims	Similarity retrieval + NLI	Evidence-linked verification	Spanish/COVID; not generative
17	IPM 2025 (C)	Misinfo detection via pretrained LMs / IPM	Thematic+sentiment+stance cues	US election tweets	RoBERTa/BERT/GPT-4 PLMs	PLMs dominate; strong F1	English; content-only
18	Eronen 2023 (C)	Cross-lingual transfer selection / IPM	Best source language, zero-shot	Language pairs	XLM-R + similarity	Similarity selection lifts transfer	No evidence/explanation
19	Skrlj 2022 (C)	KG-informed fake-news classification / Neurocomputing	Use KG knowledge	Fake-news corpora	Heterog. ensembles + KG	KG features add robustness	KG-coverage dependent
20	Kolluri 2022 (R)	COVID misinfo ML solutions / JMIR	Scale beyond human checkers	~38k articles + crowd	ML/DL + human blend	96.55% ML; 99.1% blended	Crowd on one set
21	Lu 2025 (R)	Infodemiology Multimodal attention + residual CNN / Sci. Reports	Text/image/video detection	LIAR/FakeNewsNet/WeiboNet	WeiboNet + attention fusion	0.977 acc; beat GPT-3.5	No category eval
22	Kumar 2025 (R)	Graph-augmented transformer ensemble / Sci. Reports	Accuracy + generalization	LIAR + FakeNewsNet	BERT + GNN + ensemble	96.5% acc; +4.2% F1	Heavy compute
23	Wang 2025 (R)	BERT-LSTM misinfo, mobile networks / Sci. Reports	Text-only detection	39,940 tweets	BERT + LSTM	93.51% acc	Single-modality
24	Sharma 2024 (R)	SEMTEC emotion/sentiment rumor / PeerJ CS	Emotion cues for rumor	PHEME + Twitter24	BERT + BiLSTM + TextBlob	92% on PHEME	Latency; weak real-time
25	Jadhav 2025 (R)	HEMT-Fake multilingual multimodal / Front. AI	Explainable, Indian languages	74k Hindi/Guj/Mar/Tel/Kann	XLM-R + GNN-BiLSTM + SHAP	~89% F1; +5% over XLM-R	Modest user validation
26	Feroz 2026 (R)	Urdu news with BERT+GloVe / Sci. Reports	Low-resource Urdu detection	14,178 Urdu articles	XLM-R + BERT+GloVe	F1 0.956	Text-only

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27	Almandouh 2024 (R)	Ensemble DL Arabic fake news / Sci. Reports	Arabic detection	AFND + ARABIC-FAKETWEETS	FastText + BiLSTM+BiGRU	Up to 0.99 acc	Dialect variation
28	Merzah 2026 (R)	CNN-BiLSTM Arabic & English / Front. Big Data	One model, two languages	AFND+ANS+WELFA	CNN + dual BiLSTM	94.4% / 98.9%	Satire; two languages
29	Al-alshaqi 2024 (R)	Ensemble multimodal fake news / Sensors	Text+image+video	ISOT + MediaEval2016	ML + BERT + CNN	RF 99%; +3.1% multimodal	Not global dataset
30	Alarfaj 2025 (R)	RoBERTa-Large clickbait + XAI / Sci. Reports	Explainable clickbait	32,000 headlines	RoBERTa-Large + LIME/SHAP	97% acc	English-only
31	Muqadas 2025 (R)	Sentence-embedding clickbait / Sci. Reports	Low-resource Urdu clickbait	1,000 Urdu headlines	ML + LSTM/Bi-LSTM	88% acc	No transformer
32	Mathematics 2025 (C)	ML vs DL empirical comparison / Mathematics	Benchmark models	Public benchmarks	Comparative; ALBERT	Transformers SOTA	English text
33	Mathematics 2026 (C)	Multimodal ensemble text+image / Mathematics	Fuse text+image	Public multimodal	ViLBERT + ensemble	Ensemble beats unimodal	No evidence/explanation
34	Appl. Intell. 2022 (C)	Rumor detection via GNNs / Applied Intelligence	Propagation structure	Twitter/Weibo graphs	Hierarchical GNN	Aggregation improves detection	Needs propagation graphs
35	Neural Comput. Appl. 2025 (C)	Multi-modal domain adaptation / NCAA	Reduce cross-event shift	Weibo + Twitter	BERT+EfficientNet+DA	Improves generalization	Needs event data

4. Common Findings & Patterns

1. **Two framings, one blind spot** — work splits into accuracy benchmarking and hallucination detection; almost none enforce a *guarantee* that every assertion is evidence-backed.
2. **Grounding is soft** — RAG reduces hallucination (sometimes to ~0%) but grounding is prompt-enforced, never mechanically checked.
3. **Single-model dominance** — multi-agent LLM reliability appears in journals only once or twice, and its false-agreement risk is unexamined.
4. **Cost unreported** — accuracy/F1/hallucination-rate are reported, but rarely the latency and token cost of the reliability mechanism.
5. **Same benchmarks/domains** — LIAR, FEVER, PHEME, Weibo, AFND, clinical guidelines recur; frozen-corpus reproducible benchmarking is rare.
6. **Language skew** — English dominates; Indian-multilingual and low-resource settings remain thin.

5. Recurring Limitations & Future Scope

- Soft grounding can still be violated (prompting does not prevent contradiction/over-reach beyond retrieved text).
- No structural guarantee and no explanation — verdicts are black-box labels not tied to a checkable evidence span.
- Multi-agent reliability barely studied; correlated-agreement risk under-measured.
- Latency/cost of the reliability mechanism almost never quantified.
- Domain- and language-bound; weak cross-domain/cross-lingual generalization.
- Little human-in-the-loop or deployed, end-to-end evaluation.

6. Research Gap Statement

Across the literature, LLM factual reliability is pursued either by measuring a single model’s accuracy / hallucination rate, or by **softly** grounding generation in retrieved evidence through prompting — approaches that *reduce*, but cannot *guarantee*, unsupported assertions, and that are almost never evaluated for their latency and cost. Multi-agent LLM verification, though promising, is scarcely studied in empirical journal work and carries an unexamined risk of **correlated agreement on wrong answers**. There is therefore a clear **methodological gap**: no empirical study evaluates a **hard, mechanically-verified evidence gate** — one that permits an affirmative or contradictory verdict only when it can quote a verbatim span of retrieved evidence — as the structural precondition for **multi-agent** claim verification, together with its catch-rate, false-agreement, accuracy, latency and cost trade-offs across model scales and domains.

Gap type: primarily Methodological, with a secondary Variable gap (false-agreement and cost as under-measured outcomes). This aligns the identified gap with the Aletheia system, so the gap and the eventual paper share one subject.